

SMERA RAWAL

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EDUCATION

Vellore Institute of Technology, Chennai

2024 – 2028

B.Tech in Computer Science and Data Science

Relevant Coursework: Operating Systems, DBMS, OOPs, Data Structures, Computer Networks

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Web: HTML, CSS, JavaScript

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Feature Engineering, LSTM, CNN, Transformers, Neural Networks, GenAI, Reinforcement Learning, NLP

Libraries/Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras

Tools: Git, Linux, VS Code

RESEARCH AND EXPERIENCE

Research Intern | IIT BHU | Advisor: Dr. Pankhuri Jain

May 2026 – Present

- Incoming research intern in Machine Learning and Deep Learning.
- Will work on applied AI and deep learning methodologies for research-oriented problems.

Quantitative Finance Research Consultant | WorldQuant

Apr 2026 – Present

- Developed multiple alpha signals based on market microstructure and statistical patterns in financial time-series.
- Achieved **Maximum Sharpe Ratio of 2.66, returns of over 24%** while maintaining a **drawdown less than 5%**.
- Applied statistical validation techniques including **Sharpe ratio, IC and turnover analysis**.

Learning Stellar Differential Rotation and Spot Evolution from Photometric Time Series

- Investigating differential rotation and starspot evolution from photometric light curves using deep sequence modeling.
- Rotational modulation and spot migration via **LSTM, 1D CNN and Transformers** for long temporal dependencies.
- Applying **Lomb–Scargle periodogram** for irregular time-series frequency extraction and frequency-domain splitting.
- Designing sliding-window temporal representations and evaluating robustness under synthetic noise injection.
- Interpreting learned latent representations to study astrophysical rotational dynamics and variability patterns.

PROJECTS

Astronomical Light Curve Classification using LSTM vs MLP vs 1D CNN

- Built an end-to-end *stellar time-series pipeline* with preprocessing, normalization and synthetic variability modeling.
- Implemented and compared *LSTM, MLP and 1D CNN architectures* for star classification performance benchmarking.
- Analyzed temporal dependency capture using multiple sliding-window configurations and feature scaling strategies.
- Injected *synthetic Gaussian noise* and conducted *hyperparameter tuning* to evaluate robustness.

ML-Based Drug Repurposing Pipeline (GNN + LLMs)

- Built an end-to-end pipeline combining *GNN-based binding affinity prediction* with *Tanimoto similarity* on ChEMBL.
- Modeled molecules as graphs using RDKit and trained a GIN network to predict *pChEMBL values*.
- *Integrated LangChain* for biomedical knowledge extraction and interpretable reasoning of drug–target interactions.
- Generated ranked repurposing candidates with explainable scoring and visualization.

Real-time Multimodal Deepfake Detection System

- Real-time multimodal deepfake detection pipeline by *analysis of video, audio, and lip movement streams*.
- *DL-based encoders* for facial frame analysis, synthetic speech detection, and audio–visual lip-sync consistency.
- *Transformer-based fusion architecture* to combine multimodal embeddings for deepfake probability scores.
- Support for detecting deepfake video, AI-cloned audio, audio–video mismatch, and synthetic calls.
- *Lightweight ONNX inference* for low-latency deployment and real-time detection on live WebRTC/RTP traffic.

LEADERSHIP & ACTIVITIES

Management and Outreach Member, Google Developer's Group

- Organized technical events, workshops and external collaborations to foster developer engagement.

Logistics Lead, TEDx VITC

- Managed event logistics, vendor coordination and cross-functional execution for large-scale campus events.

Sponsorships and Outreach Member, Enactus and Business Innovation Community

- Contributed to ideation sessions, startup case discussions and innovation-driven problem solving initiatives.

Content Writer, Sahayatha Club

- Created engaging and informative content for social initiatives, increasing outreach and audience engagement.

CERTIFICATIONS

WorldQuant Gold | *IQC 2026*, Generative AI Certification | *IBM*, Machine Learning | *Stanford University (Andrew Ng)*

RELEVANT ACADEMIC INTERESTS

Scientific Machine Learning, Time-Series Modeling, Stellar Variability Analysis, Representation Learning, Robust Deep Sequence Models